12/1/2025

EasyAs123 Incident Report

Network Traffic Analysis using WireShark on Kali Linux

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1. Executive Summary

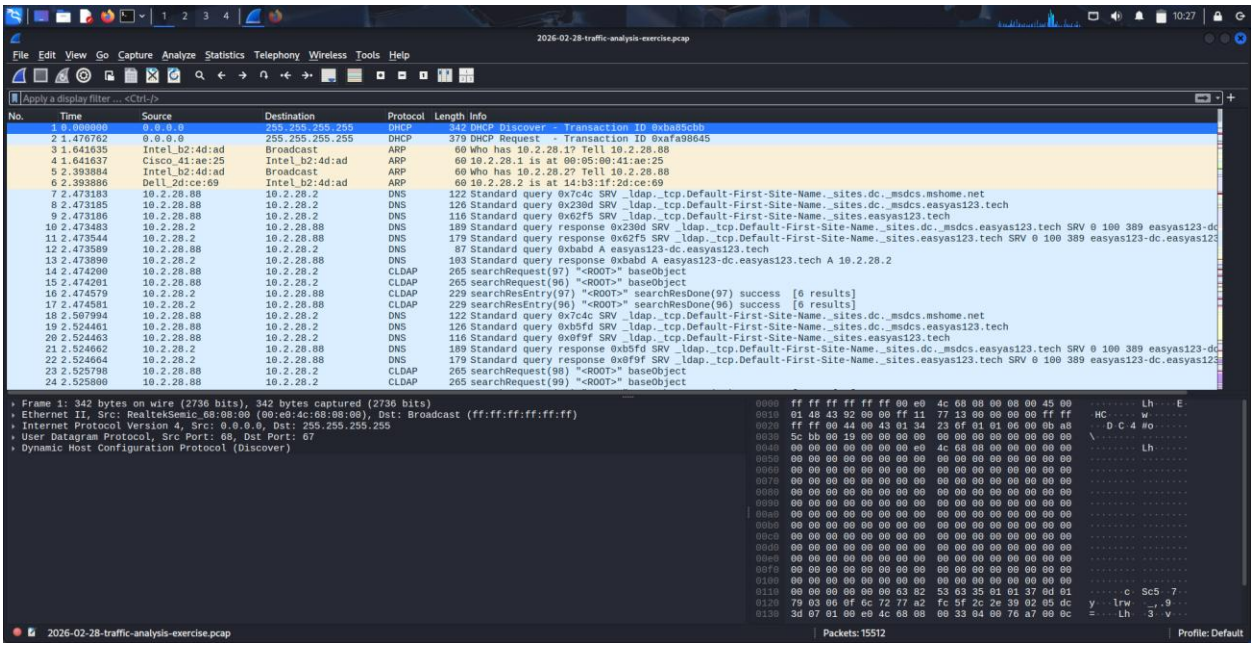
This incident report documents the findings of a network traffic analysis exercise titled "Easy As 123", published on 28 February 2026 by Malware-Traffic-Analysis.net. The analysis was conducted on Kali Linux using Wireshark as the sole forensic tool.

The exercise scenario placed the analyst in a Security Operations Center (SOC), responding to SIEM alerts for NetSupport Manager RAT activity. Multiple signature hits were recorded for outbound connections from an internal host to the external IP address 45.131.214[.]85 over TCP port 443, beginning at 19:55 UTC on 28 February 2026.

A packet capture (PCAP) file was provided for investigation. Through a structured series of Wireshark filters and built-in analysis tools, the infected Windows workstation was fully identified — including its IP address, MAC address, hostname, domain user account, and the user's full name. The full name was retrieved from a SAMR QueryUserInfo response buried within the packet data, demonstrating the depth of information recoverable from raw network traffic.

NetSupport Manager is a commercially legitimate remote administration tool that is frequently abused by threat actors to establish persistent, covert remote access to victim machines. Its use of standard ports and signed binaries often allows it to evade endpoint detection.

Exercise Title	Easy As 123 — Traffic Analysis Exercise
Source	Malware-Traffic-Analysis.net (2026-02-28)
Malware Type	NetSupport Manager RAT
C2 Server	45.131.214[.]85 — TCP port 443
Activity Start	2026-02-28 at 19:55 UTC
Domain	easyas123[.]tech	EASYAS123
Analyst Platform	Kali Linux — Wireshark



Screenshot 1 — Wireshark with full PCAP loaded; initial packet list view

2. Victim Details

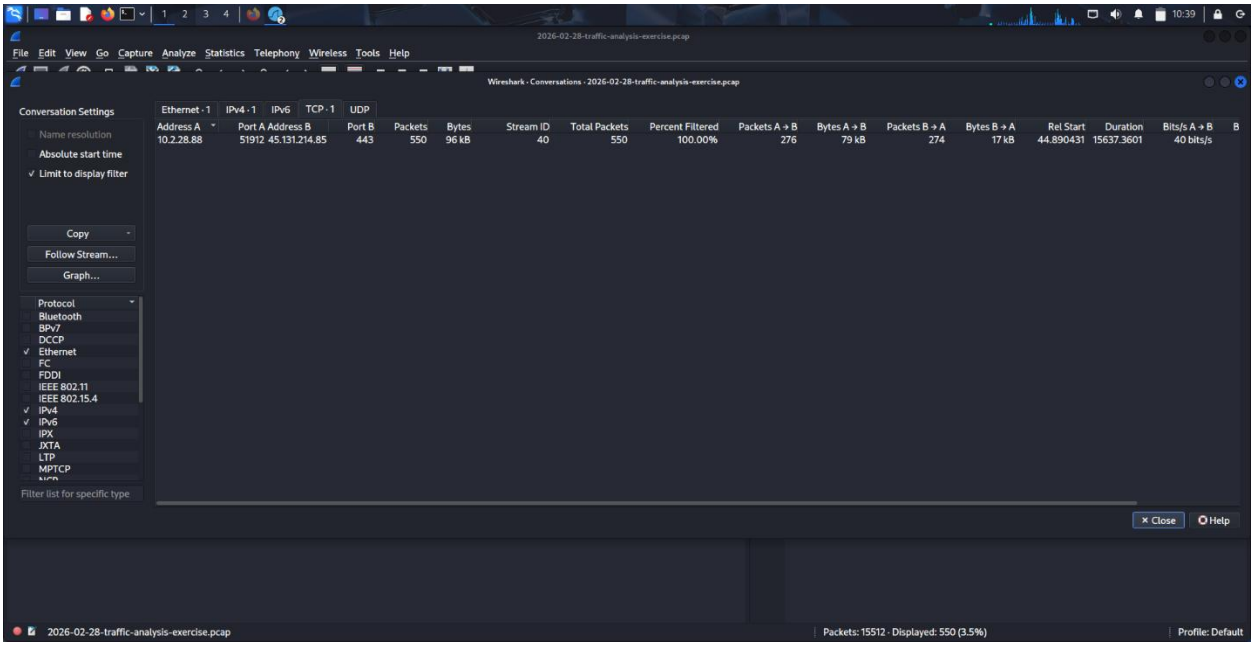
All victim details were extracted exclusively from the PCAP file using Wireshark on Kali Linux. No external tools or agents were used. The methodology for each data point is documented step-by-step below.

2.1 Identifying the Infected IP Address

The starting point was the SIEM-reported C2 server address. The following steps were taken in Wireshark:

1	Applied the display filter: ip.addr == 45.131.214.85 — this isolated all traffic to and from the known C2 server.
2	Navigated to Statistics > Conversations (TCP tab) to view all TCP sessions involving this address.
3	Identified the internal IP address in the 10.2.28.0/24 subnet that was actively communicating with 45.131.214.85. This is the infected host's IP address.

The Conversations window clearly shows the volume and direction of traffic, confirming the internal host as the source of the RAT beaconing activity.

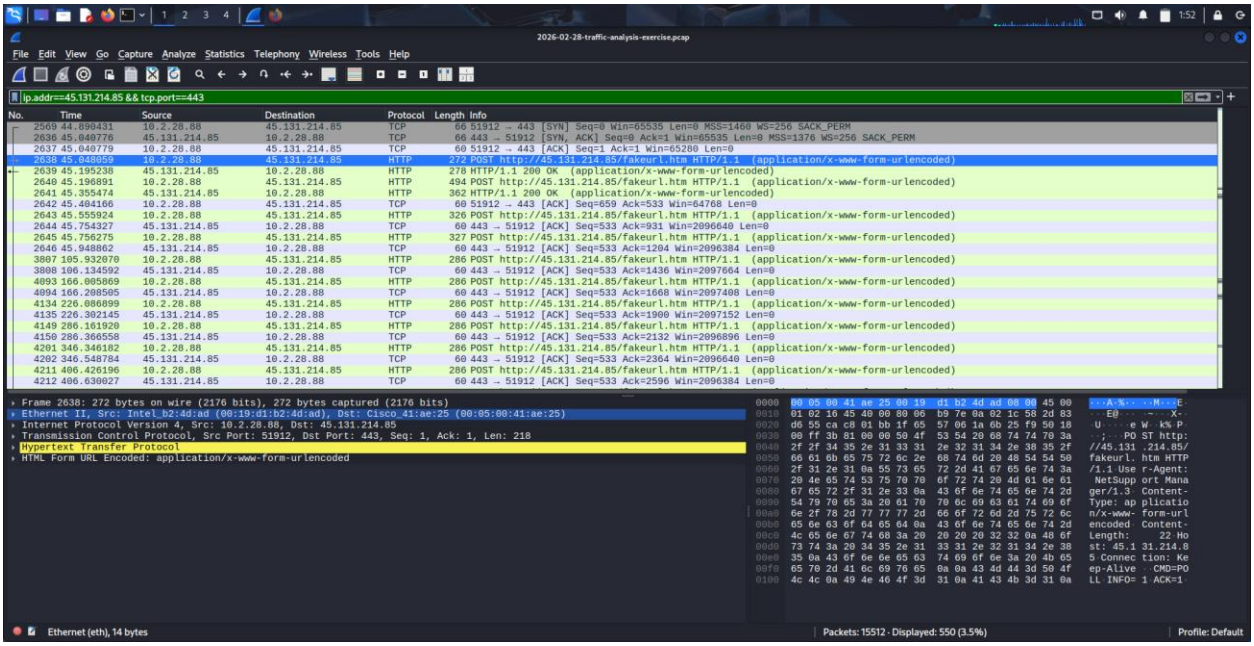


Screenshot 2 — Wireshark filter: ip.addr == 45.131.214.85; Statistics > Conversations (TCP) showing infected host IP communicating with C2

2.2 Identifying the MAC Address

Once the infected host's IP was confirmed, the MAC address was retrieved directly from the packet-level Ethernet header:

1	Cleared the display filter to show all traffic, or kept the existing filter active.
2	Clicked on any packet originating from the infected host's IP address.
3	In the Packet Details pane, expanded the Ethernet II header.
4	The Source field under the Ethernet II header displays the MAC address of the infected NIC in full.

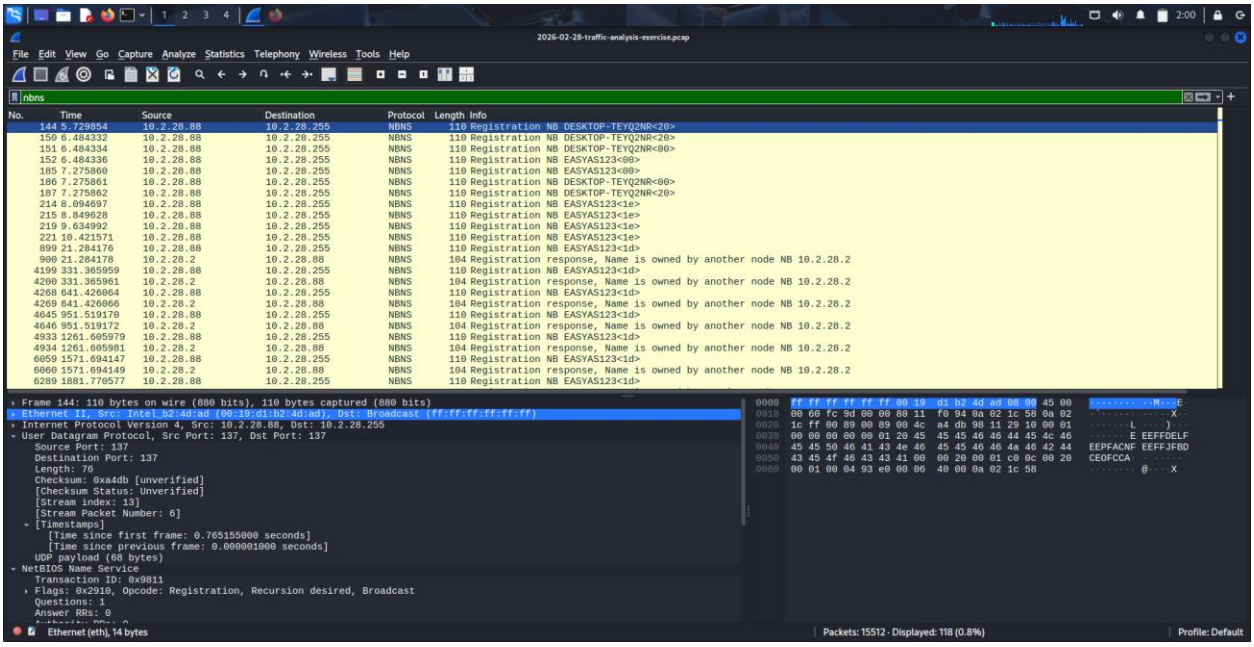


: Screenshot 3 — Packet Details pane with Ethernet II header expanded; Source MAC address of infected host highlighted

2.3 Identifying the Hostname

The NetBIOS Name Service (NBNS) protocol was used to retrieve the hostname of the infected machine. NBNS broadcasts name registration and query traffic on the local network, making it straightforward to extract hostnames from a PCAP:

1	Typed nbns in the Wireshark display filter bar and pressed Enter.
2	Reviewed the filtered NBNS packets. The Name field within NBNS registration or query packets contains the NetBIOS hostname of the broadcasting device.
3	Matched the source IP of the NBNS packet to the infected host's IP identified in Step 2.1 to confirm the hostname belongs to the correct machine.

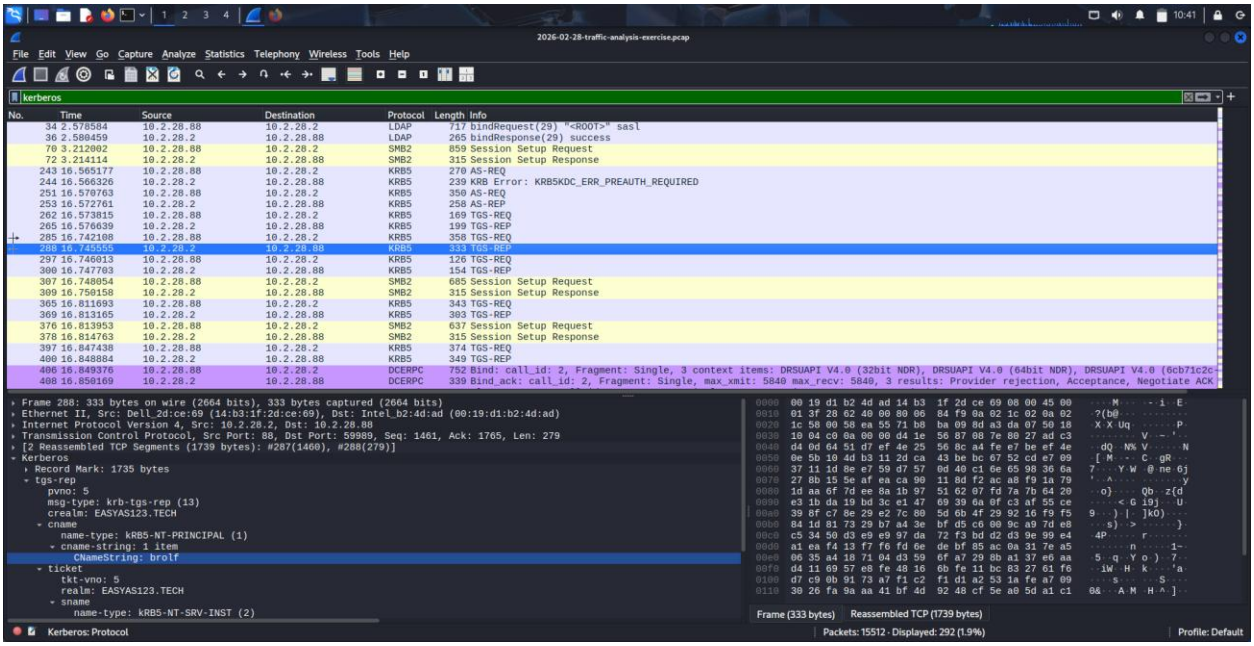


Screenshot 4 — Wireshark filter: nbns; NBNS packet from infected host IP with hostname visible in Name field

2.4 Identifying the User Account Name

The domain user account name was extracted from Kerberos authentication traffic. Kerberos is the default authentication protocol in Active Directory environments. The TGS-REP (Ticket Granting Service Reply) and AS-REP (Authentication Service Reply) packets contain the authenticated username in cleartext within their CName fields:

1	Typed kerberos in the Wireshark display filter bar and pressed Enter.
2	Browsed the filtered packets and located TGS-REP or AS-REP messages (visible in the Info column).
3	Expanded the Kerberos layer in the Packet Details pane and located the CName >CNameString field.
4	The CNameString value is the Active Directory user account name of the authenticated user on the infected host.



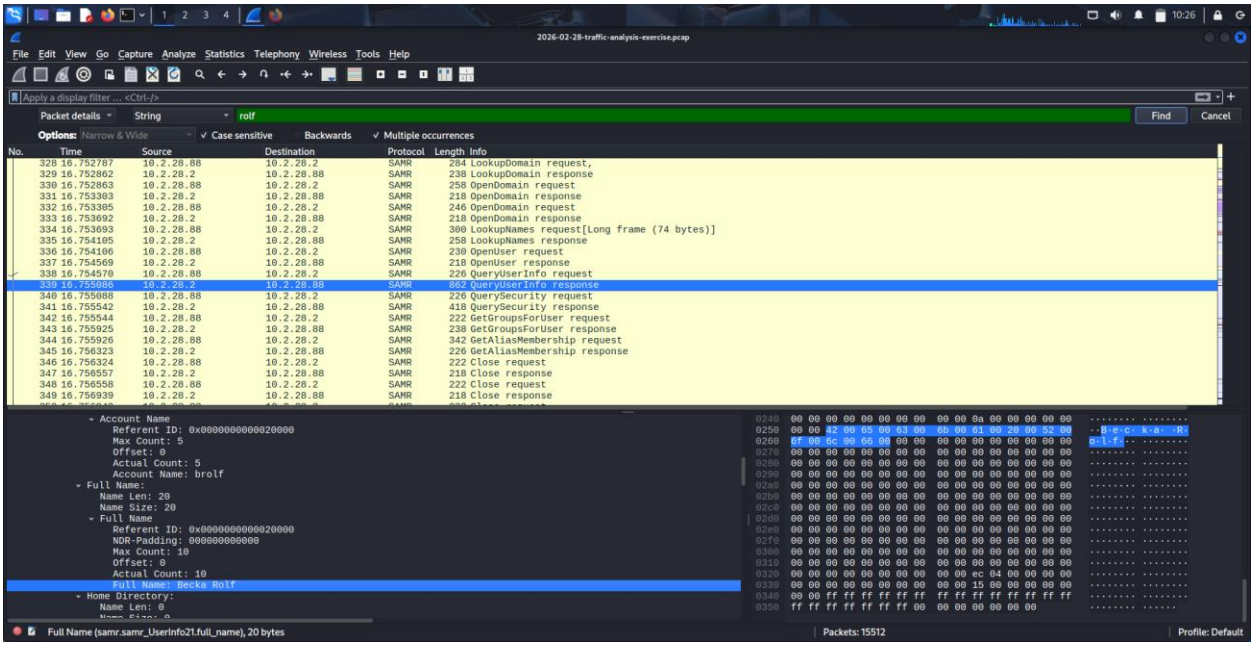
Screenshot 5 — Wireshark filter: kerberos; TGS-REP or AS-REP packet expanded showing CNameString field with username

2.5 Identifying the User's Full Name

The user's full display name was recovered from a SAMR (Security Account Manager Remote Protocol) QueryUserInfo response. This is a more advanced extraction technique that involves searching the raw packet data:

1	Cleared any active display filter.
2	Went to Edit > Find Packet (or pressed Ctrl+F) in Wireshark.
3	Set the search type to String and the search field to Packet bytes.
4	Typed rolf (the partial name used as a search string) and clicked Find.
5	Wireshark jumped to a packet containing this string within the raw payload.
6	In the Packet Details pane, expanded the SAMR (Security Account Manager Remote Protocol) layer.
7	Located the QueryUserInfo Response within the SAMR header. The Full Name field under this response contains the user's complete display name.

This technique demonstrates how sensitive account metadata — including full legal names — can be extracted from unencrypted SMB/SAMR traffic in domain environments, highlighting significant information exposure risks on poorly segmented networks.



Screenshot 6 — Edit > Find Packet with 'rolf' as search string; matching packet selected; SAMR header expanded; QueryUserInfo Response with Full Name field highlighted

2.6 Summary of Victim Details

The table below consolidates all findings about the infected host. Fill in the values discovered during your Wireshark analysis:

IP Address	10.2.28[.]88
MAC Address	00:19:d1:b2:4d:ad
Hostname	DESKTOP-TEYQ2NR
AD Domain	easyas123[.]tech
Domain Controller	10.2.28.2 — EASYAS123-DC
User Account Name	brolf
User Full Name	Becka Rolf

3. Indicators of Compromise (IOCs)

Indicators of Compromise (IOCs) are forensic artefacts that, when observed, indicate a system has likely been breached. The IOCs below were extracted from the PCAP and can be used by security teams to detect, block, or hunt for similar activity across other systems.

3.1 Network-Based IOCs

IOC Type	Value	Description
IP Address	45.131.214[.]85	External C2 server — NetSupport Manager RAT command-and-control host
Dest. Port	TCP 443	Outbound RAT traffic on HTTPS port — blends with legitimate web traffic
Protocol	NetSupport Manager	Proprietary RAT protocol tunnelled over TCP 443 to evade port-based blocking
Traffic Type	Outbound beacon	Infected host initiating persistent outbound connections to C2
First Seen	2026-02-28 19:55 UTC	Timestamp of first confirmed C2 beacon per SIEM alert and PCAP evidence

3.2 Host-Based IOCs

IOC Type	Value	Description
IP Address	10.2.28[.]88	Internal IP of the compromised Windows workstation on the 10.2.28.0/24 LAN segment
MAC Address	00:19:d1:b2:4d:ad	Physical NIC address from Ethernet II header — confirms device identity
Hostname	DESKTOP-TEYQ2NR	NetBIOS name recovered from NBNS traffic
Username	brolf	AD SAM account name from Kerberos TGS-REP / AS-REP CNameString field
Full Name	Becka Rolf	User display name from SAMR QueryUserInfo Response in raw packet bytes

3.3 About NetSupport Manager RAT

NetSupport Manager (NSM) is a commercially available remote desktop product by NetSupport Ltd. It has been co-opted by multiple threat actor groups since at least 2017 for malicious remote access because:

- It is digitally signed software, meaning many endpoint security tools treat it as trusted.
- It communicates over common ports (TCP 443, TCP 80), making network-level blocking difficult without deep packet inspection.
- Its network traffic can resemble legitimate enterprise remote support activity.
- Installation can be silently pushed via phishing emails, malicious websites, or bundled with malware loaders such as SocGholish and TA505.

Once deployed, the attacker gains full visibility of the victim's screen and keyboard, file system access, and the ability to execute arbitrary commands — all without requiring a separate exploit or vulnerability.

4. Lessons Learned and Reflections

4.1 Analytical Workflow Summary

This exercise demonstrated a clean, logical investigation workflow entirely within Wireshark, moving from a network IOC (C2 IP) to a fully identified user in five distinct steps:

1	ip.addr == 45.131.214.85 → Statistics > Conversations → Identified infected host IP on the LAN
2	Clicked any packet from infected IP → Ethernet II header → Extracted MAC address
3	nbns filter → NBNS name registration/query packets → Extracted hostname
4	kerberos filter → TGS-REP / AS-REP packets → CNameString field → Extracted username
5	Edit > Find Packet ('rolf') → SAMR QueryUserInfo Response → Full Name field → Extracted full name

Each step built on the last, using a different protocol layer to extract progressively richer identity information — from Layer 2 (MAC) all the way to application-layer AD account data (SAMR).

4.2 Key Takeaways

- A single display filter on the C2 IP is sufficient to pivot immediately to the victim host — no complex query language needed.
- The Statistics > Conversations view is a powerful triage tool for quickly visualising which internal hosts are communicating with suspicious external IPs.
- NBNS is an extremely reliable and fast way to get hostnames from Windows LAN traffic — it almost always appears in captures from Active Directory environments.
- Kerberos CNameString in TGS-REP/AS-REP is the canonical technique for extracting AD usernames from network captures without host access.
- Edit > Find Packet with a string search over Packet Bytes is an underrated technique for hunting specific values (like a known partial name) directly in the payload — useful when you don't know the exact protocol field to filter on.
- SAMR traffic in an AD domain exposes highly sensitive account metadata in cleartext, including full display names — this is a real-world security concern beyond this exercise.

4.3 Defensive Recommendations

- Block or alert on outbound connections to 45.131.214[.]85 at the perimeter firewall immediately.
- Implement deep packet inspection (DPI) on TCP 443 to detect non-TLS protocols masquerading as HTTPS, such as NetSupport Manager's proprietary protocol.
- Deploy network-based IDS/IPS (e.g., Suricata with Emerging Threats rules) to detect NetSupport Manager protocol signatures.
- Restrict installation of remote administration tools via application whitelisting (e.g., Windows Defender Application Control).
- Segment the internal network so that SAMR and NBNS traffic cannot be captured by unauthorised devices — both protocols expose sensitive information in cleartext.
- Investigate the infected host for persistence mechanisms and lateral movement before reimaging.

4.4 Personal Reflections

This is my first exercise using WireShark. Going into this exercise, I had heard of Wireshark but had never actually opened it before. The interface felt overwhelming at first as they were thousands of packets scrolling past with no obvious starting point. I was not sure where to even begin. The most challenging part for me was not any individual step in isolation, but rather connecting all the steps together into a coherent picture. Each filter, ip.addr, nbns, kerberos felt manageable on its own, but understanding how they collectively built up a full profile of the infected host took some adjustment. It required thinking not just about what each protocol does, but about why that protocol would reveal the specific piece of information I was looking for. That shift from “what button do I press” to “why does this work” was the biggest mental hurdle. The step that genuinely surprised me was retrieving the user’s full name through Edit > Find Packet and the SAMR QueryUserInfo Response. I did not expect to find a person’s real name buried in the raw bytes of a network packet. That moment made the exercise feel less like a tutorial and more like actual forensic work — the kind where you uncover something meaningful that was hiding in plain sight. This exercise has made me genuinely excited about network forensics. The idea that so much information — who was on the machine, what account they used, even their full name — can be reconstructed from a single packet capture file is both powerful and eye-opening. I intend to continue working through the exercises on Malware-Traffic-Analysis.net and to keep building my Wireshark skills, as this experience has shown me just how much depth there is to explore in this field.

5. References and Appendix

5.1 References

[1] Brad Duncan. (2026, February 28). Traffic Analysis Exercise: Easy As 123. Malware-Traffic-Analysis.net.
https://www.malware-traffic-analysis.net/2026/02/28/index.html

[2] NetSupport Ltd. NetSupport Manager — Remote PC Management Software.
https://www.netsupportsoftware.com/netsupport-manager/

[3] Wireshark Foundation. Wireshark User's Guide — Display Filters.
https://www.wireshark.org/docs/wsug_html/

[4] Microsoft. [MS-SAMR]: Security Account Manager (SAM) Remote Protocol Specification.
https://learn.microsoft.com/en-us/openspecs/windows_protocols/ms-samr/

[5] Kali Linux. Official Documentation.
https://www.kali.org/docs/

5.2 Appendix A — Wireshark Filters and Techniques Used

Filter / Technique	Purpose
ip.addr == 45.131.214.85	Isolate all traffic to/from the C2 server
Statistics > Conversations (TCP)	Identify internal IP communicating with C2
Click packet > Ethernet II header	Extract MAC address of infected host
nbns	Retrieve hostname from NBNS name broadcasts
kerberos	Filter Kerberos auth traffic for TGS-REP/AS-REP
Kerberos CNameString field	Extract AD user account (SAM) name
Edit > Find Packet > String 'rolf' in Bytes	Search raw payload for partial username string
SAMR > QueryUserInfo Response > Full Name	Extract user's full display name from SAMR traffic

5.3 Appendix B — Environment Details

Analyst OS	Kali Linux
Analysis Tool	Wireshark
PCAP File	2026-02-28-traffic-analysis-exercise.pcap
PCAP Size	5.2 MB (5,167,384 bytes)
LAN Segment	10.2.28.0/24 (10.2.28.0 – 10.2.28.255)
Domain	easyas123[.]tech
AD Name	EASYAS123
Domain Controller	10.2.28.2 — EASYAS123-DC
LAN Gateway	10.2.28.1
LAN Broadcast	10.2.28.255

5.4 Appendix C — Screenshot Index

Screenshot 1	Wireshark — Full PCAP loaded, initial packet list (Executive Summary)
Screenshot 2	Wireshark — ip.addr==45.131.214.85 filter + Statistics > Conversations (TCP), infected IP visible (Section 2.1)
Screenshot 3	Wireshark — Ethernet II header expanded, Source MAC address of infected host (Section 2.2)
Screenshot 4	Wireshark — nbns filter, NBNS packet from infected IP with hostname in Name field (Section 2.3)
Screenshot 5	Wireshark — kerberos filter, TGS-REP/AS-REP packet expanded with CNameString username (Section 2.4)
Screenshot 6	Wireshark — Edit > Find Packet with 'rolf' as search string; matching packet selected + SAMR header expanded; QueryUserInfo Response with Full Name field highlighted (Section 2.5)

— End of Report —